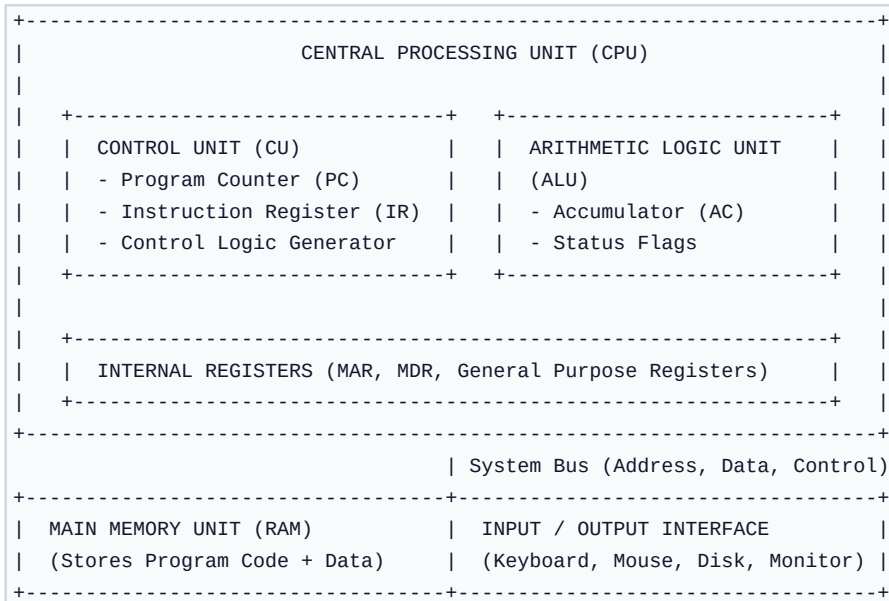


10-Mark Questions & Answers -- MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Explain in detail the Von Neumann model of computer architecture. Also discuss the block-level description of functional units (CU, ALU, Memory, I/O) with a diagram.

1. Introduction to Von Neumann Architecture



2. Detailed Functional Units

1. Control Unit (CU)

Q2. Describe all the addressing modes used in the 8086 family with suitable examples for each. Explain how each mode calculates the effective address.

1. Effective Address (EA) Calculation Formula

2. Detailed 8086 Addressing Modes

Addressing Mode	Effective Address Formula	Example Instruction	EA & Physical Address Explanation
Immediate	No EA (Operand in code).	<code>MOV AX, 5000H</code>	Data <code>5000H</code> is encoded directly inside the instruction.

Register Direct	No EA (Operand in register).	`MOV AX, BX`	Data taken directly from CPU register
Direct Memory	**EA = Displacement**	`MOV AX, [2000H]`	**EA = 2000H** , **PA = (DS x 16) + 200
Register Indirect	**EA = [BX/BP/SI/DI]**	`MOV AX, [BX]`	If `BX = 1500H`, **EA = 1500H** .
Based	**EA = Base + Disp**	`MOV AX, [BX + 08H]`	If `BX = 1000H`, **EA = 1000H + 08H =
Indexed	**EA = Index + Disp**	`MOV AX, [SI + 10H]`	If `SI = 2000H`, **EA = 2000H + 10H =
Based-Indexed	**EA = Base + Index**	`MOV AX, [BX + SI]`	If `BX = 1000H, SI = 0050H`, **EA = 10
Based-Indexed with Disp	**EA = Base + Index + Disp**	`MOV AX, [BX + SI + 04H]`	If `BX = 1000H, SI = 0050H`, **EA = 10

Q3. Explain the evolution of computers from the first to the fifth generation, highlighting key technological changes and performance improvements at each stage.

COMPUTER GENERATIONS EVOLUTION	
1st Gen (1940-56)	: Vacuum Tubes ----> 1000s of Operations/sec
2nd Gen (1956-63)	: Transistors -----> 100,000s Operations/sec
3rd Gen (1964-71)	: Integrated Circuits (SSI/MSI) -> Millions Ops/sec
4th Gen (1971-Pres)	: Microprocessors (VLSI/ULSI) -> Billions Ops/sec
5th Gen (Future)	: Artificial Intelligence & Quantum Computing

1. 1st Generation (1940-1956): Vacuum Tubes

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Q4. Discuss various performance metrics used to evaluate computer architecture with formulas, and explain Amdahl's Law with a numerical example.

1. Performance Metrics & Formulas

1. CPU Execution Time (T_{CPU}):

2. Cycles Per Instruction (CPI):

3. Millions of Instructions Per Second (MIPS):

T_C

CPI

MIP

4. Floating-Point Operations Per Second (FLOPS):

FLO

2. Amdahl's Law

Amdah

$$\text{Speedup_Overall} = 1 / [(1 - f) + (f / s)]$$

3. Numerical Example

* P
matrix

* Result: The overall system execution speed increases by 2.78 times (or a 178% speedup).